

Detailed Notes on Prototyping in HCI

HCI - Week 2, Fall 2025

Abstract

These notes cover the fundamental concepts of prototyping within the field of Human-Computer Interaction (HCI). The document begins by defining user interface prototyping and its role in the design process, then explores the continuum from low-fidelity sketches to high-fidelity interactive models. It provides a detailed comparison of different fidelity levels, including their respective advantages and disadvantages. A significant portion is dedicated to the psychological principles of perception, including perceptual biases and Gestalt principles, which form the theoretical foundation for effective interface design. Finally, the notes survey the modern landscape of prototyping, including advanced tools for AI-assisted design, voice and immersive experiences, remote testing, and accessibility.

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1 Introduction to User Interface Prototyping (Slides 1-2)

1.1 The Role of Prototyping in HCI

The development of highly interactive software systems with Graphical User Interfaces (GUIs) is now standard. The success and user acceptance of these systems depend heavily on the quality of the GUI. A user-friendly interface can be the deciding factor between a successful product and a failed one.

What is Prototyping?

Prototyping is an essential practice in User Interface (UI) design. It involves creating a preliminary model or simulation of a final product. It serves as an excellent means of generating and exploring design ideas, helping to evaluate the quality of a solution at an early and inexpensive stage.

By creating prototypes, designers can:

- Test and refine design concepts with users before committing to development.
- Communicate design ideas more effectively to stakeholders (e.g., developers, clients, management).
- Identify and fix usability issues early in the design process, which is significantly cheaper than fixing them post-launch.

2 The Sketch to Design Continuum (Slides 3-8)

The design process is not a single step but a progression from abstract ideas to a fully realized product. This journey can be visualized as a continuum, moving from low commitment and low fidelity to high commitment and high fidelity.

The key stages in this continuum are:

2.1 Sketch

- **Definition:** Sketching is a form of rapid, freehand drawing with *no intention* of it becoming a finished product.
- **Purpose:** It is a tool for exploration and ideation. Sketches are about generating a high quantity of ideas, exploring possibilities, and communicating concepts quickly. They are meant to be disposable.
- **Key Idea:** Sketching supports the *process of making*, not the actual design itself. It is the foundational step upon which more refined design work is built.

2.2 Wireframe

- **Definition:** A wireframe is a basic visual guide, often called a blueprint or a skeleton, that suggests the structure of an interface and the relationships between its pages or components.
- **Purpose:** Wireframes are used to communicate and explore the concepts that emerge from sketching. They focus on:
 - **Information Architecture:** How information is organized and presented.
 - **Structure and Layout:** Where elements are placed on the screen.
 - **Overall Flow:** How a user navigates through the interface to complete tasks.
- **Important Note:** Wireframes are created *before* any detailed visual design. They typically lack color, specific typography, or actual imagery, focusing purely on structure.

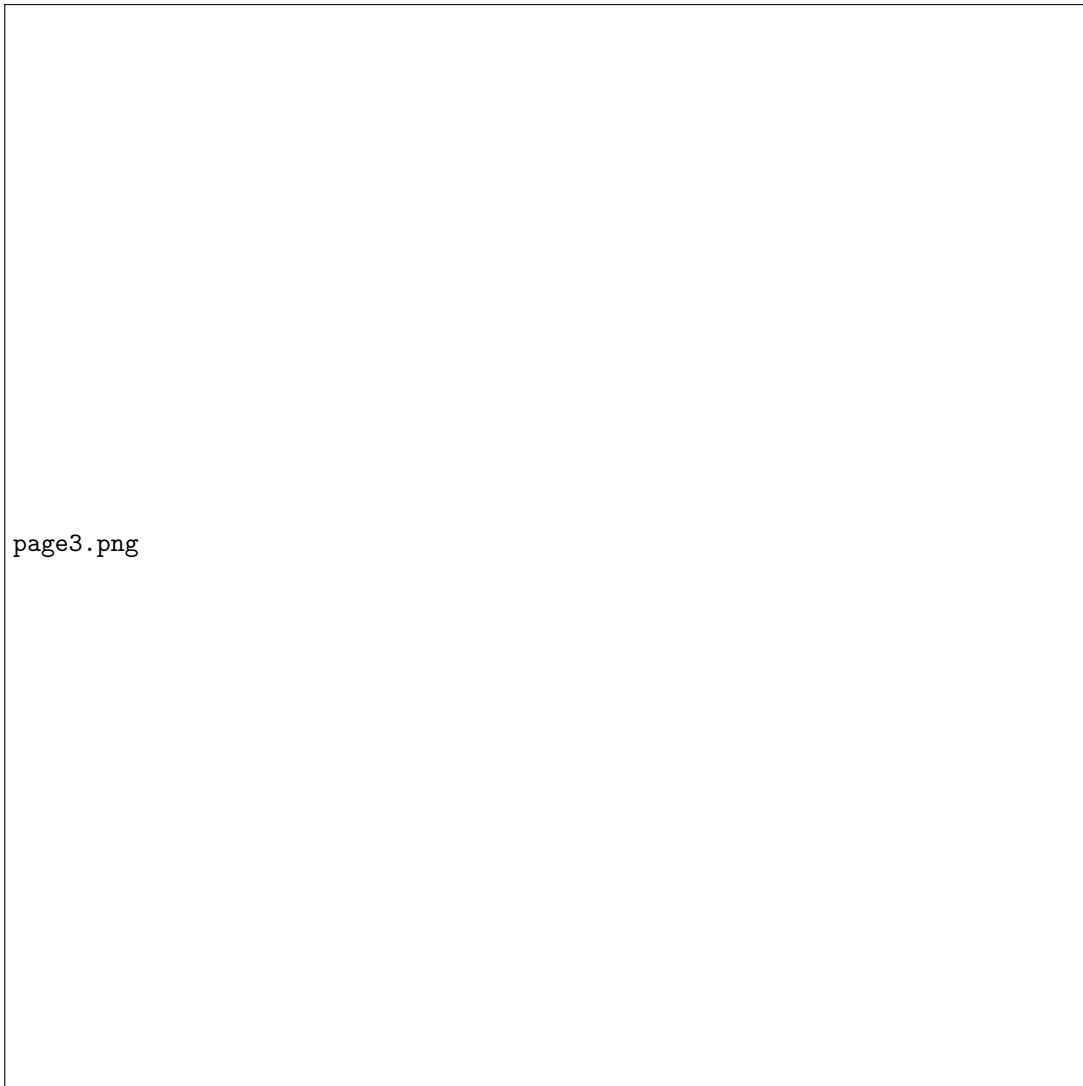


Figure 1: The Sketch to Design Continuum, illustrating the iterative process from ideation to a final product.

2.3 Prototype

- **Definition:** A prototype is an expression of design intent that simulates the final interaction between the user and the interface. It's a more advanced model than a wireframe because it incorporates interactivity.
- **Purpose:** Prototyping allows designers and users to "see the design in action." Depending on its purpose, a prototype can simulate a single interaction (like a button click) or the workflow of an entire application.

2.4 Progression Summary

The journey from concept to product can be summarized in three phases, as shown in the visual from slide 8:

1. **IDEATE (Sketch):** Brainstorming and exploring ideas.
2. **DETALIZE (Wireframe):** Adding structure, flow, and detail to the concepts.
3. **VISUALIZE (Prototype):** Creating an interactive and visual representation of the final product.

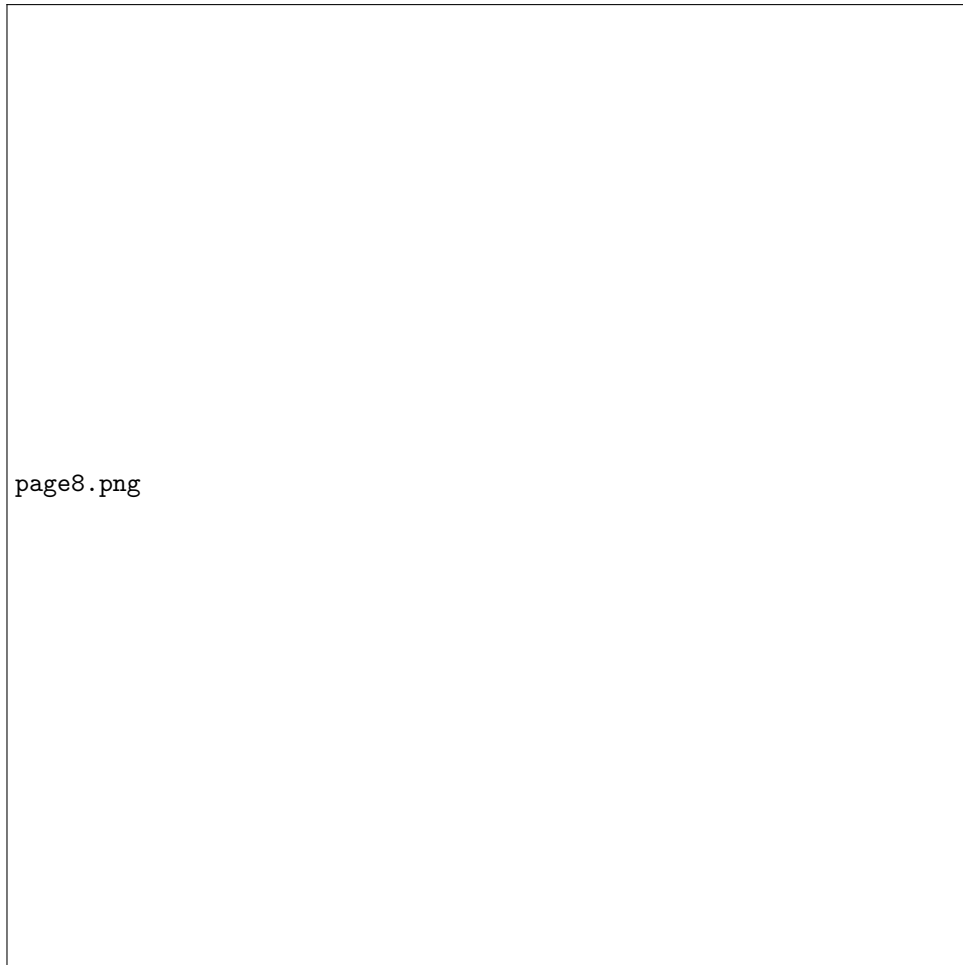


Figure 2: The progression from ideation to visualization.

3 Fidelity in Prototyping (Slides 9-19)

3.1 What is Fidelity?

The **fidelity** of a prototype refers to its level of detail and realism compared to the final product. Prototypes are not always polished and can exist anywhere on a spectrum from low to high fidelity.

Fidelity can vary across three main dimensions:

- **Visual Design:** How closely it resembles the final look and feel (e.g., colors, typography, spacing).
- **Content:** Whether it uses placeholder text (e.g., "Lorem Ipsum") or real, final content.
- **Interactivity:** The extent to which a user can interact with it (e.g., clickable buttons, animations, data entry).

Teams choose a prototype's fidelity based on their goals, the completeness of the design, and available resources (time, money, skills). The two extremes are:

- **Low-Fidelity (Lo-Fi)**
- **High-Fidelity (Hi-Fi)**

3.2 Low-Fidelity Prototyping

- **Description:** Lo-Fi prototypes are simple, often paper-based, and do not typically allow for direct user interaction. They use a medium different from the final product, such as paper, cardboard, or a whiteboard.

- **Characteristics:** They are quick, cheap, and easy to change.
- **Advantage:** Because they look rough and unfinished, users often feel more comfortable providing honest feedback and suggesting major changes. They are less intimidated than when presented with a polished design.
- **Example:** A series of hand-drawn mockups of a mobile app's screens, where a facilitator manually switches the drawings in response to the user's "taps." This is often called a **paper prototype**.

3.2.1 Storyboarding

Storyboarding is a common low-fidelity technique.

- It is a series of sketches or drawings that visualizes how a user might progress through a task using a device or service.
- It is often used with *scenarios* (narrative descriptions of user goals) to bring more detail and context to the design.
- It is an excellent tool for role-playing and understanding the user's journey in a holistic way.

3.3 High-Fidelity Prototyping

- **Description:** Hi-Fi prototypes are computer-based and closely resemble the final product in both look and feel. They are fully interactive systems.
- **Characteristics:** They allow for realistic user interactions (e.g., mouse clicks, keyboard entry). Users can enter data, respond to messages, and navigate as if it were a real system.
- **Purpose:** They are highly effective for:
 - Collecting true human performance data (e.g., time to complete a task, error rates).
 - Demonstrating the product to clients, management, and other stakeholders in a convincing way.

3.4 Advantages and Disadvantages of Fidelity Levels

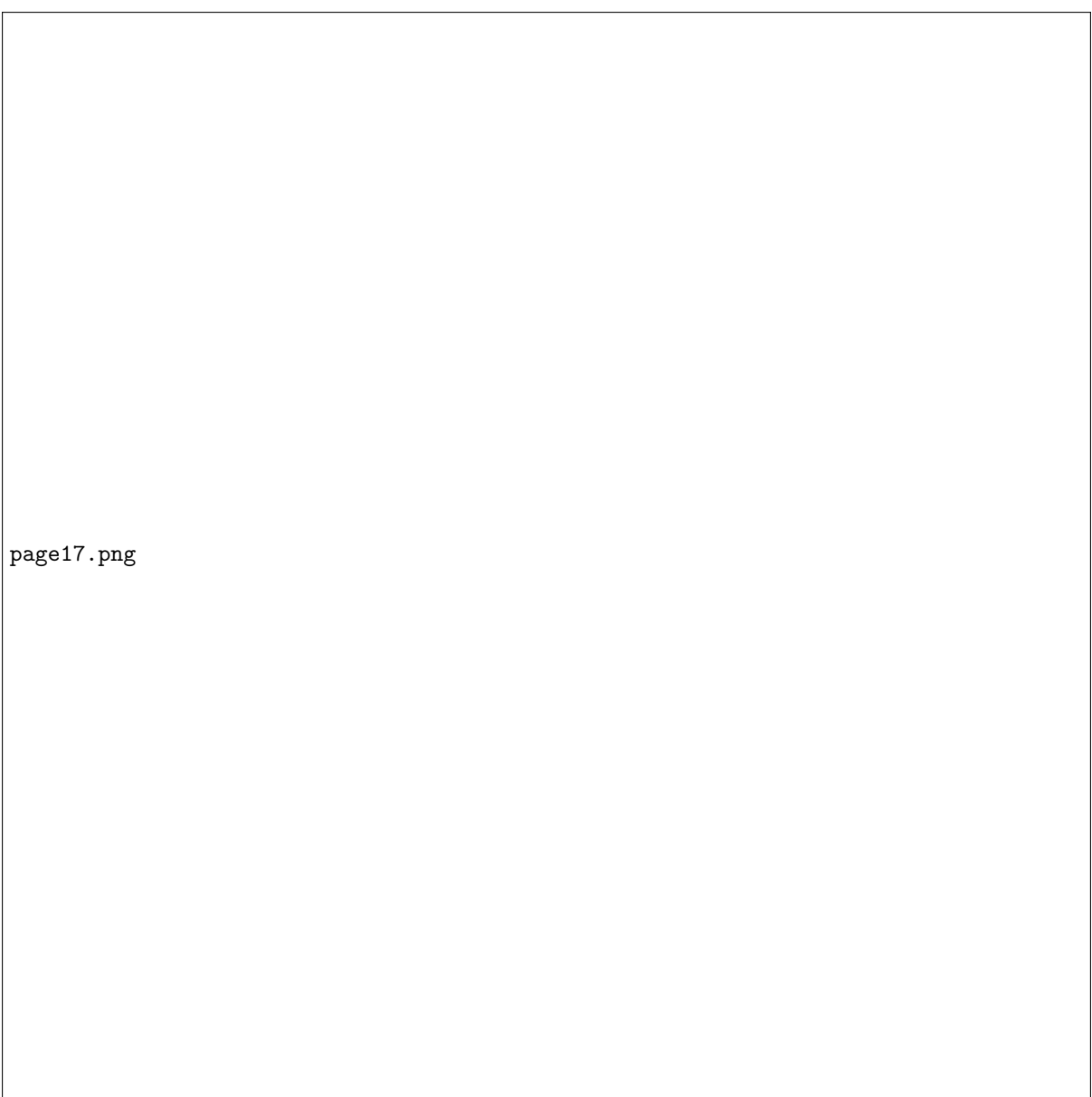
3.4.1 Low-Fidelity

Advantages

- **Quick and Cheap:** Can be created in minutes with simple materials.
- **Encourages Iteration:** Low investment makes it easy to discard ideas and try new ones.
- **Focus on High-Level Concepts:** Allows designers and users to focus on information architecture and flow, rather than getting distracted by visual details like color or font choice.
- **More User Feedback:** Users are less hesitant to criticize a rough sketch.

Disadvantages

- **May Seem Unprofessional:** Some stakeholders might find it difficult to look past the rough nature.
- **Lacks Realism:** Does not accurately represent the user experience of the final product.
- **Limited Navigation:** Requires a facilitator (a human "computer") to "make the interaction happen," which can be cumbersome.



page17.png

Figure 3: Example of a high-fidelity prototype for a mobile coffee ordering application.

3.4.2 High-Fidelity

Advantages

- **Realistic Interactions:** Provides a much better sense of the final user experience.
- **Better for Testing:** More effective for collecting quantitative performance data.
- **User-Driven:** Users can interact with the prototype on their own, without a facilitator.
- **More Persuasive:** Highly effective for demonstrations to stakeholders.

Disadvantages

- **Time and Skill Intensive:** Takes significantly longer to build and requires specialized skills and tools.
- **Reluctance to Change:** After investing significant effort, designers may become resistant to making changes (the "sunk cost" fallacy).
- **Surface-Level Feedback:** Testers and reviewers may focus on minor surface details (e.g., "I don't like that shade of blue") rather than on core functionality and structure.

4 The Psychology of Perception in Design

To create effective prototypes and designs, we must first understand how the human mind perceives the world. Our perception is not a perfect reflection of reality; it is actively constructed and biased by our experiences, context, and goals.

4.1 We Perceive What We Expect

Our brain constantly makes predictions about the world. These expectations are shaped by three key factors:

- **The Past:** Our prior experiences.
- **The Present:** The current context.
- **The Future:** Our current goals.

4.1.1 Perception Biased by Experience

Our past experiences "prime" our perceptual system to see things in a certain way.

- **Example 1: Buildings vs. Word:** When shown an image of five black shapes, people who are told they are looking at a map of buildings will see buildings. If they are told they are looking at a word, they will clearly see the word "LIFT" in the white space. The physical stimulus is the same, but the perception is entirely different based on priming.
- **Example 2: The Dalmatian:** An image that initially looks like a random collection of black spots becomes a clear picture of a Dalmatian dog sniffing the ground once you are told what to look for. Afterward, it is nearly impossible to go back to seeing it as just random spots.

4.1.2 Perception Biased by Context

The surrounding environment or context influences how we interpret information.

- **Example: THE CHT:** A character that is physically identical is perceived as an 'H' in the word 'THE' and as an 'A' in the word 'CAT'. The context provided by the surrounding letters determines our perception.

4.1.3 Perception Biased by Goals

Our current goals act as a filter, causing us to notice things that are relevant to our goal and ignore things that are not. This phenomenon is known as **inattentional blindness**.

- **Example: Website Navigation:** A user tasked with finding a campus map on a university website will scan the page for links like "Maps" or "Departments." They are highly likely to completely ignore a banner on the same page announcing they have won \$100, because it is unrelated to their goal.

Design Implications of Perceptual Bias

1. **Avoid Ambiguity:** Design interfaces that are clear and can be interpreted in only one way.
2. **Be Consistent:** Place controls and information that serve the same function in the same location on every page. Consistency in layout, color, and typography allows users to rely on their experience and recognize elements quickly without conscious effort.
3. **Understand User Goals:** Designers must understand what users are trying to achieve. Information needed for a specific goal should be prominent and clearly available at every step of the interaction.

4.2 Gestalt Principles of Visual Perception

Early in the 20th century, German psychologists described a set of principles explaining how our visual system automatically imposes structure on visual input. These are known as the **Gestalt principles**. The core idea is that we perceive whole shapes and figures rather than disconnected parts. These principles are fundamental to graphic and interface design.

4.2.1 Principle of Proximity

Objects that are close to each other are perceived as a group.

- **Application:** In UI design, related controls (like an "Add" and "Remove" button for a list) should be placed close together, and separated by space from unrelated controls (like "OK" and "Cancel" for the whole window).

4.2.2 Principle of Similarity

Objects that look similar (in shape, color, size, etc.) are perceived as a group.

- **Application:** Making all clickable links the same color helps users perceive them as a related group of interactive elements.

4.2.3 Principle of Continuity

The mind prefers to follow continuous paths and lines rather than abrupt changes. We perceive continuous forms over disconnected segments.

- **Application:** The famous IBM logo is perceived as three letters, even though it is made of disconnected horizontal stripes, because our mind follows the continuous path of each letterform.

4.2.4 Principle of Closure

Our visual system automatically tries to close open figures to see them as whole objects.

- **Application:** An icon showing a stack of documents only needs to show the full top document and the edges of the ones behind it. Our brain fills in the gaps and perceives a full stack.

4.2.5 Principle of Symmetry

We tend to parse complex scenes into simpler, symmetrical shapes.

- **Application:** A complex shape is more likely to be interpreted as two simple, overlapping, symmetrical diamonds rather than two asymmetrical shapes touching at a corner.

4.2.6 Principle of Figure/Ground

Our mind separates the visual field into a **figure** (the object of our attention) and a **ground** (the background).

- **Application:** Modal windows or "pop-ups" use this principle. The pop-up becomes the figure, and the rest of the page is often dimmed to become the ground, focusing the user's attention.

4.2.7 Principle of Common Fate

Objects that move in the same direction are perceived as a single group.

- **Application:** When multiple files are selected on a desktop and dragged together, their shared motion reinforces that they are being treated as a single group.

5 The Modern Prototyping Landscape (Slides 21-29)

Prototyping has evolved significantly with the advent of sophisticated digital tools.

5.1 Interactive Prototyping Tools

Modern tools have revolutionized the prototyping process.

- **Examples:** Figma, Adobe XD, Sketch, InVision.
- **Key Features:** These tools enable the creation of highly interactive and realistic prototypes, live collaboration between team members (similar to Google Docs), and seamless handoff to developers (providing them with design specifications, assets, and even code snippets).

5.2 AI-Assisted Prototyping

AI is accelerating the early stages of design.

- **Functionality:** AI-powered tools can generate wireframes, mockups, and even entire design layouts from simple text prompts or hand-drawn sketches.
- **Examples:**
 - **Galileo AI:** Generates UI designs from plain English descriptions.
 - **Uizard:** Converts hand-drawn sketches into editable digital prototypes.

5.3 Code-Linked Prototypes

These tools bridge the gap between design and development.

- **Functionality:** They allow designers to create prototypes that are so high-fidelity they can export production-ready code.
- **Examples:**
 - **Framer:** Creates interactive prototypes using React components that can be used directly in production.
 - **FlutterFlow:** A visual builder that generates code for the Flutter framework.

5.4 Emerging Frontiers in Prototyping

Prototyping is expanding beyond traditional graphical interfaces.

- **Voice & Conversational Prototyping:** Tools for designing and testing voice interfaces (VUIs) and chatbots.
 - *Examples:* Voiceflow (for Alexa, Google Assistant), Botmock.
- **AR/VR & Spatial Prototyping:** Tools for designing immersive augmented and virtual reality experiences.
 - *Examples:* Adobe Aero (no-code AR), ShapesXR (collaborative VR design).

5.5 Integrating Testing and Analytics

Modern workflows integrate feedback and data directly into the prototyping process.

- **Remote Usability Testing:** Platforms that connect with prototypes to gather user feedback from real users remotely.
 - *Examples:* Maze, UserTesting.
- **Analytics-Integrated Prototyping:** Tools that provide behavioral analytics like heatmaps, click-tracking, and user journey maps within the prototype itself.
 - *Examples:* Hotjar, FullStory.

5.6 Accessibility & Inclusivity

Ensuring products are usable by everyone, including those with disabilities, is a critical part of modern design. Prototyping tools are increasingly incorporating accessibility features.

- **Functionality:** These tools help designers check for issues like poor color contrast and screen reader incompatibility early in the process.
- **Examples:**
 - **Stark plugin for Figma:** Checks for color contrast and other accessibility compliance issues.
 - **Axe DevTools:** Tests prototypes for screen reader compatibility.

6 Conclusion and Future of Prototyping

6.1 Conclusion

- Lo-fi prototypes are invaluable for their low cost and ease of iteration, allowing teams to focus on core interaction design and information architecture.
- Computer-based prototypes are essential for detailed testing, electronic distribution, and documenting the design process.
- The choice of prototype method is a strategic decision. Often, the best approach is to **combine several techniques**, starting with lo-fi methods for exploration and moving to hi-fi methods for refinement and validation.

6.2 The Future of Prototyping

Prototyping is evolving to become more integrated, intelligent, and realistic.

- **Trend:** The future is smart, collaborative, and AI-powered simulations that closely mirror real-world use.
- **Benefits:** This evolution reduces development costs, speeds up iteration cycles, and ultimately enhances user satisfaction.
- **Example Workflow:** Combining an AI generation tool like Galileo with a testing platform like Maze provides a powerful, end-to-end workflow for designing and validating products efficiently.